

Mathematical and Physical Modeling of Entangled Photon Propagation in Optical Fibers

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Chapter 1

Purpose of This Document

This document collects detailed mathematical derivations supporting the theoretical and numerical development of the quantum fiber simulator.

The derivations reported here are not intended to interrupt the main thesis narrative. Their role is to provide a rigorous analytical reference for results that are used in the manuscript, in the MATLAB implementation, and in the validation of numerical experiments.

The main thesis text contains the physical motivation, final mathematical results, and physical interpretation. Detailed algebraic passages, intermediate expansions, and reusable analytical validations are instead collected here.

Whenever a derivation becomes essential for reader-facing reproducibility, it may later be adapted into the thesis appendices.

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Chapter 2

Density Operators and Reduced States

2.1 Expectation Values in the Density-Operator Formalism

This derivation shows how the density-operator expression for expectation values generalizes the pure-state expression.

For a pure state $|\psi\rangle$, the expectation value of an observable O is

$$\langle O \rangle = \langle \psi | O | \psi \rangle. \quad (2.1)$$

The associated density operator is

$$\rho = |\psi\rangle\langle\psi|. \quad (2.2)$$

Using this definition, the density-operator expression becomes

$$\langle O \rangle = \text{Tr}(\rho O) = \text{Tr}(|\psi\rangle\langle\psi| O). \quad (2.3)$$

To show that this reproduces the pure-state expression, expand the trace in an arbitrary orthonormal basis $\{|k\rangle\}$:

$$\begin{aligned} \text{Tr}(\rho O) &= \sum_k \langle k | \psi \rangle \langle \psi | O | k \rangle \\ &= \sum_k \langle \psi | O | k \rangle \langle k | \psi \rangle \\ &= \langle \psi | O \left(\sum_k |k\rangle\langle k| \right) | \psi \rangle \\ &= \langle \psi | O | \psi \rangle. \end{aligned} \quad (2.4)$$

Therefore,

$$\boxed{\langle O \rangle = \text{Tr}(\rho O)} \quad (2.5)$$

reproduces the standard pure-state expectation value whenever $\rho = |\psi\rangle\langle\psi|$.

The advantage of the density-operator formalism becomes evident when the system is not described by a single pure state, but by a statistical ensemble. Suppose the system is prepared in one of the pure states $|\psi_k\rangle$ with probability p_k , where

$$p_k \geq 0, \quad \sum_k p_k = 1.$$

The density operator of the ensemble is

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|. \quad (2.6)$$

The expectation value of O is then the statistical average of the corresponding pure-state expectation values:

$$\begin{aligned} \langle O \rangle &= \sum_k p_k \langle \psi_k | O | \psi_k \rangle \\ &= \sum_k p_k \text{Tr} (|\psi_k\rangle\langle\psi_k| O) \\ &= \text{Tr} \left[\left(\sum_k p_k |\psi_k\rangle\langle\psi_k| \right) O \right] \\ &= \text{Tr}(\rho O). \end{aligned} \quad (2.7)$$

Thus, the same expression

$$\langle O \rangle = \text{Tr}(\rho O) \quad (2.8)$$

applies both to pure states and to statistical mixtures.

This result is central for the simulator because it allows observable quantities to be evaluated through a unified mathematical interface, independently of whether the quantum state is represented as a pure state vector or as a density operator.

In particular, correlation observables of the form $\sigma_i \otimes \sigma_j$, local polarization observables, and state diagnostics can all be computed through the same trace expression once the corresponding operator has been constructed.

The density-operator expectation value therefore provides the mathematical bridge between deterministic pure-state evolution, ensemble-averaged descriptions, and effective non-unitary models arising from decoherence or statistical uncertainty.

2.2 Partial Trace for Maximally Entangled States

This derivation establishes explicitly that the reduced density operators of a maximally entangled Bell state are maximally mixed.

Consider the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle). \quad (2.9)$$

The corresponding density operator is

$$\rho_{AB} = |\Psi^+\rangle\langle\Psi^+|. \quad (2.10)$$

Expanding explicitly:

$$\begin{aligned} \rho_{AB} &= \frac{1}{2} (|01\rangle + |10\rangle) (\langle 01| + \langle 10|) \\ &= \frac{1}{2} (|01\rangle\langle 01| + |01\rangle\langle 10| + |10\rangle\langle 01| + |10\rangle\langle 10|). \end{aligned} \quad (2.11)$$

The reduced density operator of subsystem A is obtained by tracing over subsystem B :

$$\rho_A = \text{Tr}_B(\rho_{AB}). \quad (2.12)$$

Using the computational basis $\{|0\rangle, |1\rangle\}$ for subsystem B , the partial trace can be written as

$$\rho_A = \sum_{b=0}^1 (I_A \otimes \langle b|) \rho_{AB} (I_A \otimes |b\rangle). \quad (2.13)$$

Each contribution in Eq. (2.11) can now be evaluated separately.

For the first diagonal term:

$$\begin{aligned} \text{Tr}_B (|01\rangle\langle 01|) &= |0\rangle\langle 0| \langle 1|1\rangle \\ &= |0\rangle\langle 0|. \end{aligned} \quad (2.14)$$

Similarly,

$$\text{Tr}_B (|10\rangle\langle 10|) = |1\rangle\langle 1|. \quad (2.15)$$

The off-diagonal terms vanish because of basis orthogonality:

$$\begin{aligned} \text{Tr}_B (|01\rangle\langle 10|) &= |0\rangle\langle 1| \langle 1|0\rangle \\ &= 0, \end{aligned} \quad (2.16)$$

and analogously,

$$\text{Tr}_B (|10\rangle\langle 01|) = 0. \quad (2.17)$$

Collecting all contributions gives

$$\begin{aligned} \rho_A &= \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) \\ &= \frac{1}{2} I. \end{aligned} \quad (2.18)$$

By symmetry, the same result holds for subsystem B :

$$\rho_B = \frac{1}{2}I. \tag{2.19}$$

This result has a direct physical interpretation.

Although the global bipartite state $|\Psi^+\rangle$ is pure, each subsystem individually is described by a maximally mixed state.

Consequently, no local measurement performed on a single subsystem can reveal complete information about the global quantum state.

The information characterizing the entangled state is therefore encoded entirely in the nonlocal correlations between the two subsystems.

This property constitutes one of the fundamental signatures of quantum entanglement and establishes the conceptual connection between entanglement and local mixedness explored throughout the present work. ““

Chapter 3

Phase-Shifted Bell-State Formalism

3.1 Metrics for Phase-Shifted Bell States

This derivation collects the analytical state metrics associated with the phase-shifted Bell-state family and with its random-phase ensemble extension.

The deterministic phase-shifted Bell state is defined as

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} (|01\rangle + e^{i\theta}|10\rangle). \quad (3.1)$$

The corresponding density operator is

$$\begin{aligned} \rho(\theta) &= |\psi(\theta)\rangle\langle\psi(\theta)| \\ &= \frac{1}{2}|01\rangle\langle 01| + \frac{1}{2}|10\rangle\langle 10| + \frac{e^{-i\theta}}{2}|01\rangle\langle 10| + \frac{e^{i\theta}}{2}|10\rangle\langle 01|. \end{aligned} \quad (3.2)$$

In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, this density operator is represented by

$$\rho(\theta) = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & e^{-i\theta} & 0 \\ 0 & e^{i\theta} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (3.3)$$

3.1.1 Reduced Density Operators

The reduced density operators are obtained by tracing over the complementary subsystem:

$$\rho_A(\theta) = \text{Tr}_B[\rho(\theta)], \quad \rho_B(\theta) = \text{Tr}_A[\rho(\theta)]. \quad (3.4)$$

Using Eq. (3.2), the diagonal terms give

$$\text{Tr}_B(|01\rangle\langle 01|) = |0\rangle\langle 0|, \quad \text{Tr}_B(|10\rangle\langle 10|) = |1\rangle\langle 1|, \quad (3.5)$$

whereas the off-diagonal terms vanish under the partial trace:

$$\text{Tr}_B(|01\rangle\langle 10|) = 0, \quad \text{Tr}_B(|10\rangle\langle 01|) = 0. \quad (3.6)$$

Therefore,

$$\rho_A(\theta) = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I. \quad (3.7)$$

The same result holds for subsystem B :

$$\rho_B(\theta) = \frac{1}{2}I. \quad (3.8)$$

Thus, the relative phase modifies the global coherence terms but leaves both local states maximally mixed.

3.1.2 Deterministic Phase Metrics

Since $\rho(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|$, the global state is pure for every value of θ :

$$\gamma(\rho(\theta)) = \text{Tr}(\rho(\theta)^2) = 1. \quad (3.9)$$

The reduced purities are instead

$$\gamma(\rho_A(\theta)) = \gamma(\rho_B(\theta)) = \text{Tr} \left[\left(\frac{I}{2} \right)^2 \right] = \frac{1}{2}. \quad (3.10)$$

The fidelity with respect to the reference Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

is

$$\begin{aligned} F_{\Psi^+}(\rho(\theta)) &= |\langle\Psi^+|\psi(\theta)\rangle|^2 \\ &= \left| \frac{1}{2} (1 + e^{i\theta}) \right|^2 \\ &= \frac{1 + \cos \theta}{2}. \end{aligned} \quad (3.11)$$

For the pure state

$$|\psi(\theta)\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle,$$

with

$$a = 0, \quad b = \frac{1}{\sqrt{2}}, \quad c = \frac{e^{i\theta}}{\sqrt{2}}, \quad d = 0,$$

the concurrence is

$$\begin{aligned} C(|\psi(\theta)\rangle) &= 2|ad - bc| \\ &= 2 \left| -\frac{e^{i\theta}}{2} \right| \\ &= 1. \end{aligned} \quad (3.12)$$

Hence, deterministic phase evolution changes the overlap with the fixed reference Bell state but does not reduce either global purity or entanglement.

3.1.3 Random-Phase Ensemble Density Operator

Consider an ensemble of phase-shifted Bell states $|\psi(\theta_k)\rangle$, each occurring with probability p_k . The ensemble density operator is

$$\rho_{AB}^{(\text{ens})} = \sum_k p_k |\psi(\theta_k)\rangle \langle \psi(\theta_k)|, \quad p_k \geq 0, \quad \sum_k p_k = 1. \quad (3.13)$$

Define the ensemble coherence parameter

$$\mu = \langle e^{i\theta} \rangle = \sum_k p_k e^{i\theta_k}. \quad (3.14)$$

Equivalently,

$$\mu = \langle \cos \theta \rangle + i \langle \sin \theta \rangle, \quad (3.15)$$

where

$$\langle \cos \theta \rangle = \sum_k p_k \cos \theta_k, \quad \langle \sin \theta \rangle = \sum_k p_k \sin \theta_k. \quad (3.16)$$

Using Eq. (3.2), the ensemble state becomes

$$\rho_{AB}^{(\text{ens})} = \frac{1}{2} |01\rangle \langle 01| + \frac{1}{2} |10\rangle \langle 10| + \frac{\mu^*}{2} |01\rangle \langle 10| + \frac{\mu}{2} |10\rangle \langle 01|. \quad (3.17)$$

In matrix form,

$$\rho_{AB}^{(\text{ens})} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \mu^* & 0 \\ 0 & \mu & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (3.18)$$

The deterministic state is recovered when $\mu = e^{i\theta}$, while complete phase averaging corresponds to $\mu = 0$.

3.1.4 Ensemble Reduced States and Purity

By linearity of the partial trace, the ensemble reduced states remain maximally mixed:

$$\rho_A^{(\text{ens})} = \rho_B^{(\text{ens})} = \frac{1}{2} I. \quad (3.19)$$

The global purity is obtained from Eq. (3.18). Since only the two-dimensional subspace spanned by $|01\rangle$ and $|10\rangle$ contributes, one may square the nonzero block:

$$\left[\frac{1}{2} \begin{pmatrix} 1 & \mu^* \\ \mu & 1 \end{pmatrix} \right]^2 = \frac{1}{4} \begin{pmatrix} 1 + |\mu|^2 & 2\mu^* \\ 2\mu & 1 + |\mu|^2 \end{pmatrix}. \quad (3.20)$$

Taking the trace gives

$$\gamma(\rho_{AB}^{(\text{ens})}) = \text{Tr} \left[\left(\rho_{AB}^{(\text{ens})} \right)^2 \right] = \frac{1 + |\mu|^2}{2}. \quad (3.21)$$

Using Eq. (3.15), this can be written as

$$\gamma\left(\rho_{AB}^{(\text{ens})}\right) = \frac{1 + \langle \cos \theta \rangle^2 + \langle \sin \theta \rangle^2}{2}. \quad (3.22)$$

The reduced purities are independent of μ :

$$\gamma\left(\rho_A^{(\text{ens})}\right) = \gamma\left(\rho_B^{(\text{ens})}\right) = \frac{1}{2}. \quad (3.23)$$

3.1.5 Ensemble Fidelity with Respect to $|\Psi^+\rangle$

The fidelity with respect to $|\Psi^+\rangle$ is

$$\begin{aligned} F_{\Psi^+}\left(\rho_{AB}^{(\text{ens})}\right) &= \langle \Psi^+ | \rho_{AB}^{(\text{ens})} | \Psi^+ \rangle \\ &= \frac{1}{2} (1 + \text{Re}(\mu)). \end{aligned} \quad (3.24)$$

Since $\text{Re}(\mu) = \langle \cos \theta \rangle$, this becomes

$$F_{\Psi^+}\left(\rho_{AB}^{(\text{ens})}\right) = \frac{1 + \langle \cos \theta \rangle}{2}. \quad (3.25)$$

Therefore, fidelity with respect to $|\Psi^+\rangle$ depends only on the real part of the coherence parameter, not on its full magnitude.

3.1.6 Ensemble Concurrence

The random-phase ensemble state has the two-qubit X-state structure. For this family, the only nonzero anti-diagonal coherence relevant to concurrence is the coherence between $|01\rangle$ and $|10\rangle$. The concurrence is therefore

$$C\left(\rho_{AB}^{(\text{ens})}\right) = |\mu|. \quad (3.26)$$

In trigonometric form,

$$C\left(\rho_{AB}^{(\text{ens})}\right) = \sqrt{\langle \cos \theta \rangle^2 + \langle \sin \theta \rangle^2}. \quad (3.27)$$

Thus, concurrence depends on the magnitude of the ensemble coherence parameter and is insensitive to its phase orientation in the complex plane.

3.1.7 Summary of Metric Relations

For the deterministic phase-shifted Bell state $|\psi(\theta)\rangle$, the relevant metrics are

$$\gamma(\rho(\theta)) = 1, \quad \gamma(\rho_A(\theta)) = \gamma(\rho_B(\theta)) = \frac{1}{2}, \quad C(|\psi(\theta)\rangle) = 1, \quad (3.28)$$

and

$$F_{\Psi^+}(\rho(\theta)) = \frac{1 + \cos \theta}{2}. \quad (3.29)$$

For the random-phase ensemble, the metrics are controlled by

$$\mu = \langle e^{i\theta} \rangle = \langle \cos \theta \rangle + i \langle \sin \theta \rangle. \quad (3.30)$$

The corresponding relations are

$$\rho_A^{(\text{ens})} = \rho_B^{(\text{ens})} = \frac{1}{2}I, \quad (3.31)$$

$$\gamma \left(\rho_{AB}^{(\text{ens})} \right) = \frac{1 + |\mu|^2}{2} = \frac{1 + \langle \cos \theta \rangle^2 + \langle \sin \theta \rangle^2}{2}, \quad (3.32)$$

$$F_{\Psi^+} \left(\rho_{AB}^{(\text{ens})} \right) = \frac{1 + \text{Re}(\mu)}{2} = \frac{1 + \langle \cos \theta \rangle}{2}, \quad (3.33)$$

and

$$C \left(\rho_{AB}^{(\text{ens})} \right) = |\mu| = \sqrt{\langle \cos \theta \rangle^2 + \langle \sin \theta \rangle^2}. \quad (3.34)$$

These expressions show that the ensemble coherence parameter μ provides a compact analytical description of coherence loss in the random-phase model. Its magnitude controls both global purity and concurrence, while its real part controls the fidelity with respect to the fixed reference Bell state $|\Psi^+\rangle$.

Chapter 4

Correlation Tensor and Measurement Formalism

4.1 Correlation Tensor of the Phase-Evolved Bell State

This derivation computes explicitly the correlation tensor associated with the reduced phase model.

The phase-evolved Bell state is

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} (|01\rangle + e^{i\theta}|10\rangle). \quad (4.1)$$

The corresponding density operator is

$$\rho(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|. \quad (4.2)$$

Expanding explicitly gives

$$\begin{aligned} \rho(\theta) &= \frac{1}{2} (|01\rangle + e^{i\theta}|10\rangle) (\langle 01| + e^{-i\theta}\langle 10|) \\ &= \frac{1}{2} (|01\rangle\langle 01| + e^{-i\theta}|01\rangle\langle 10| + e^{i\theta}|10\rangle\langle 01| + |10\rangle\langle 10|). \end{aligned} \quad (4.3)$$

The correlation tensor is defined by

$$T_{ij}(\theta) = \text{Tr} [\rho(\theta) (\sigma_i \otimes \sigma_j)], \quad i, j \in \{x, y, z\}. \quad (4.4)$$

Equivalently, since the state is pure,

$$T_{ij}(\theta) = \langle\psi(\theta)|\sigma_i \otimes \sigma_j|\psi(\theta)\rangle. \quad (4.5)$$

The Pauli matrices act on the computational basis as

$$\begin{aligned} \sigma_x|0\rangle &= |1\rangle, & \sigma_x|1\rangle &= |0\rangle, \\ \sigma_y|0\rangle &= i|1\rangle, & \sigma_y|1\rangle &= -i|0\rangle, \\ \sigma_z|0\rangle &= |0\rangle, & \sigma_z|1\rangle &= -|1\rangle. \end{aligned} \quad (4.6)$$

We now compute each nonzero tensor component.

Diagonal components

For T_{xx} , one has

$$(\sigma_x \otimes \sigma_x)|01\rangle = |10\rangle, \quad (\sigma_x \otimes \sigma_x)|10\rangle = |01\rangle.$$

Therefore,

$$\begin{aligned} T_{xx}(\theta) &= \frac{1}{2} (\langle 01| + e^{-i\theta} \langle 10|) (|10\rangle + e^{i\theta} |01\rangle) \\ &= \frac{1}{2} (e^{i\theta} + e^{-i\theta}) \\ &= \cos \theta. \end{aligned} \tag{4.7}$$

For T_{yy} , one has

$$(\sigma_y \otimes \sigma_y)|01\rangle = |10\rangle, \quad (\sigma_y \otimes \sigma_y)|10\rangle = |01\rangle.$$

Hence the same calculation gives

$$T_{yy}(\theta) = \cos \theta. \tag{4.8}$$

For T_{zz} , one has

$$(\sigma_z \otimes \sigma_z)|01\rangle = -|01\rangle, \quad (\sigma_z \otimes \sigma_z)|10\rangle = -|10\rangle.$$

Thus,

$$\begin{aligned} T_{zz}(\theta) &= \langle \psi(\theta) | \sigma_z \otimes \sigma_z | \psi(\theta) \rangle \\ &= -1. \end{aligned} \tag{4.9}$$

Off-diagonal transverse components

For T_{xy} , one has

$$(\sigma_x \otimes \sigma_y)|01\rangle = -i|10\rangle, \quad (\sigma_x \otimes \sigma_y)|10\rangle = i|01\rangle.$$

Therefore,

$$\begin{aligned} T_{xy}(\theta) &= \frac{1}{2} (\langle 01| + e^{-i\theta} \langle 10|) (-i|10\rangle + ie^{i\theta} |01\rangle) \\ &= \frac{1}{2} (ie^{i\theta} - ie^{-i\theta}) \\ &= -\sin \theta. \end{aligned} \tag{4.10}$$

For T_{yx} , one has

$$(\sigma_y \otimes \sigma_x)|01\rangle = i|10\rangle, \quad (\sigma_y \otimes \sigma_x)|10\rangle = -i|01\rangle.$$

Therefore,

$$\begin{aligned}
T_{yx}(\theta) &= \frac{1}{2} (\langle 01| + e^{-i\theta} \langle 10|) (i|10\rangle - ie^{i\theta}|01\rangle) \\
&= \frac{1}{2} (-ie^{i\theta} + ie^{-i\theta}) \\
&= \sin \theta.
\end{aligned} \tag{4.11}$$

Vanishing components

All remaining components involving one transverse Pauli operator and one z -Pauli operator vanish:

$$T_{xz} = T_{zx} = T_{yz} = T_{zy} = 0. \tag{4.12}$$

This follows because these operators map the subspace spanned by $|01\rangle$ and $|10\rangle$ to the orthogonal subspace spanned by $|00\rangle$ and $|11\rangle$, or produce contributions with zero overlap against $|\psi(\theta)\rangle$.

Collecting the computed components gives

$$T(\theta) = \begin{pmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{4.13}$$

The result shows that the reduced phase θ acts as a rotation of the transverse correlation components in the x - y plane, while the longitudinal anticorrelation $T_{zz} = -1$ remains unchanged.

This is consistent with the physical interpretation of the reduced phase model: the local phase shift modifies coherence between $|01\rangle$ and $|10\rangle$, but does not alter their populations.